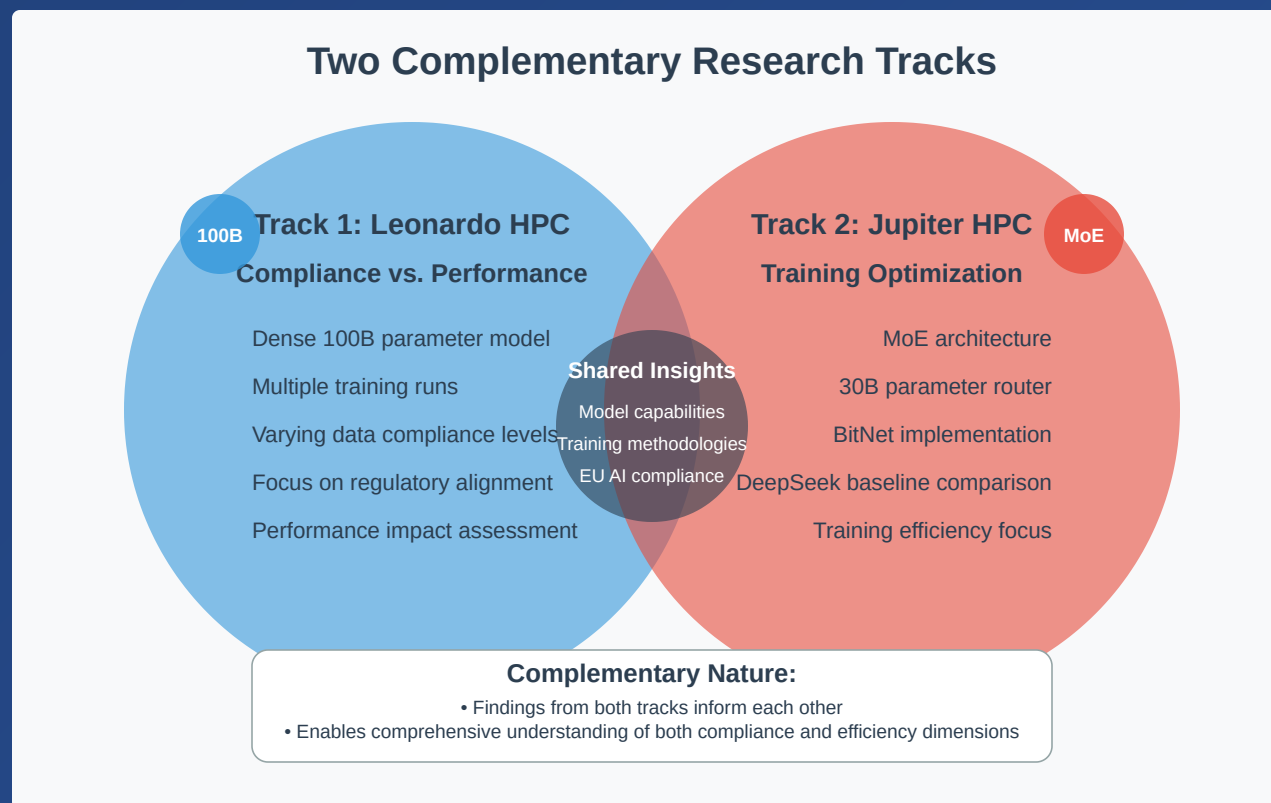


TrustLLM Training Project: Dual-Track Approach to Compliant Language Models

Dual Research Focus: Compliance-Performance Balance & Advanced Training Optimization



Project Overview

- **Objective:** Develop large language models that comply with European regulations while understanding compliance-performance tradeoffs
- **Research Focus:** Dual-track approach using complementary computing resources
- **Compliance Emphasis:** Exploring methods to balance regulatory requirements with model capabilities
- **Long-term Vision:** Building foundation for future compliant AI systems in Europe
- **Partners:** Collaboration across multiple EU research institutions and technical teams

Compute Resources Status

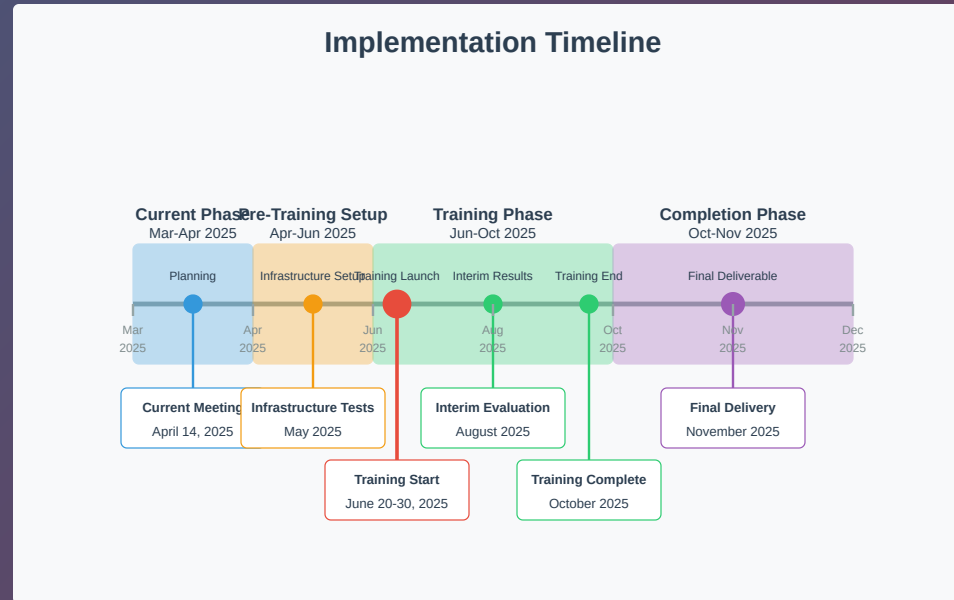
Leonardo BOOSTER (CONFIRMED)

- Large-scale training on Leonardo Booster system at CINECA (Bologna)
- Total Storage: 240 TB
- Training: ~3M A100 node hours => ~12M core-hours
- Allocation confirmed for 100B parameter model training

Jupiter EuroHPC (AWAITING CONFIRMATION)

- Applied for compute allocation: ~5M H100-hours
- Allocation sufficient for:
 - - 15T training tokens for 34B parameter model
 - - 15T training tokens for 8B parameter model
- Timeframe: Response expected in April

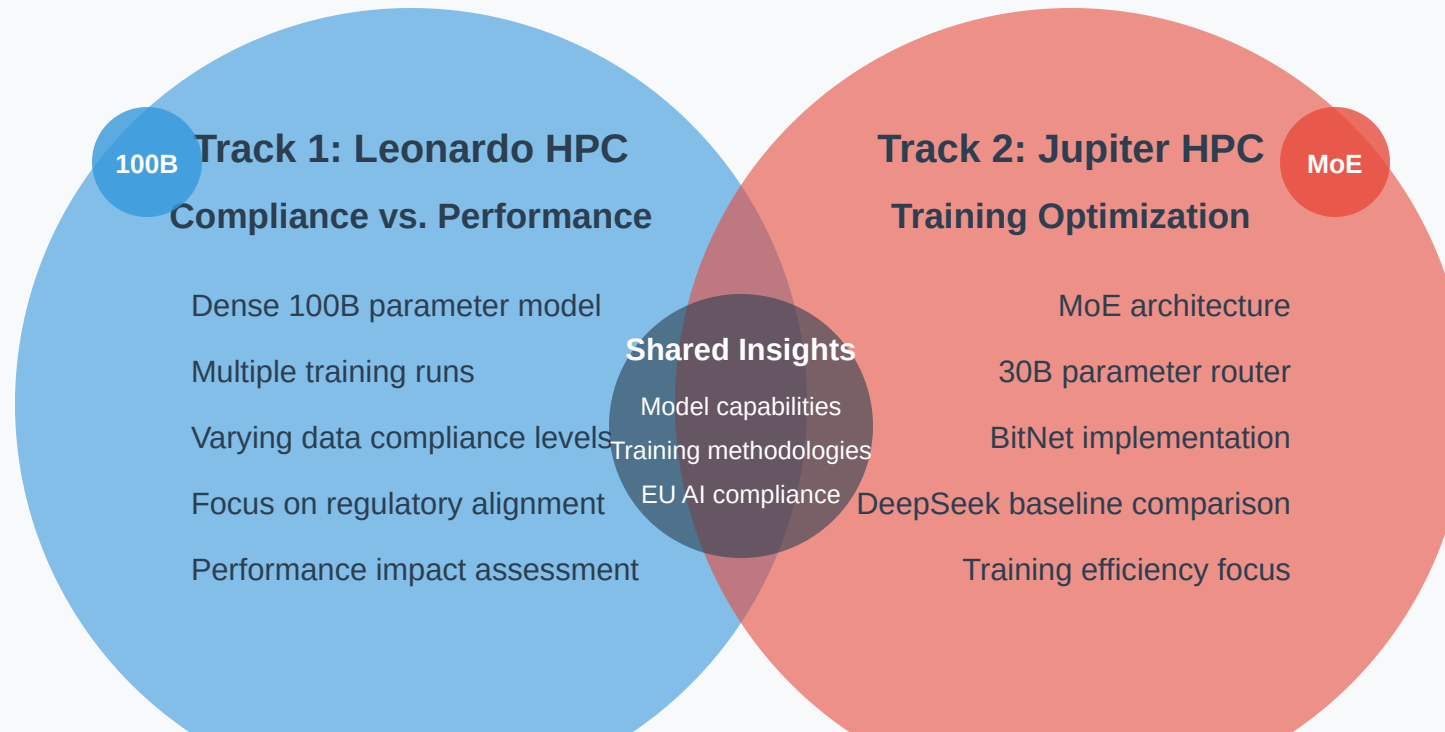
Project Roadmap



- **Current Phase** (March-April 2025): Pre-Training preparation with current meeting April 14th, 2025
- **Pre-Training Setup** (April-June 2025): Final infrastructure preparation and verification
- **Training Commencement** (June 20-30, 2025): Launch of both Leonardo and Jupiter training runs
- **Training Duration** (June-October 2025): 2-4 month training period for both models
- **Interim Evaluation** (August 2025): Analysis of preliminary results
- **Final Model Delivery** (November 2025): Completed models with compliance-performance documentation

Two Complementary Research Tracks

Two Complementary Research Tracks

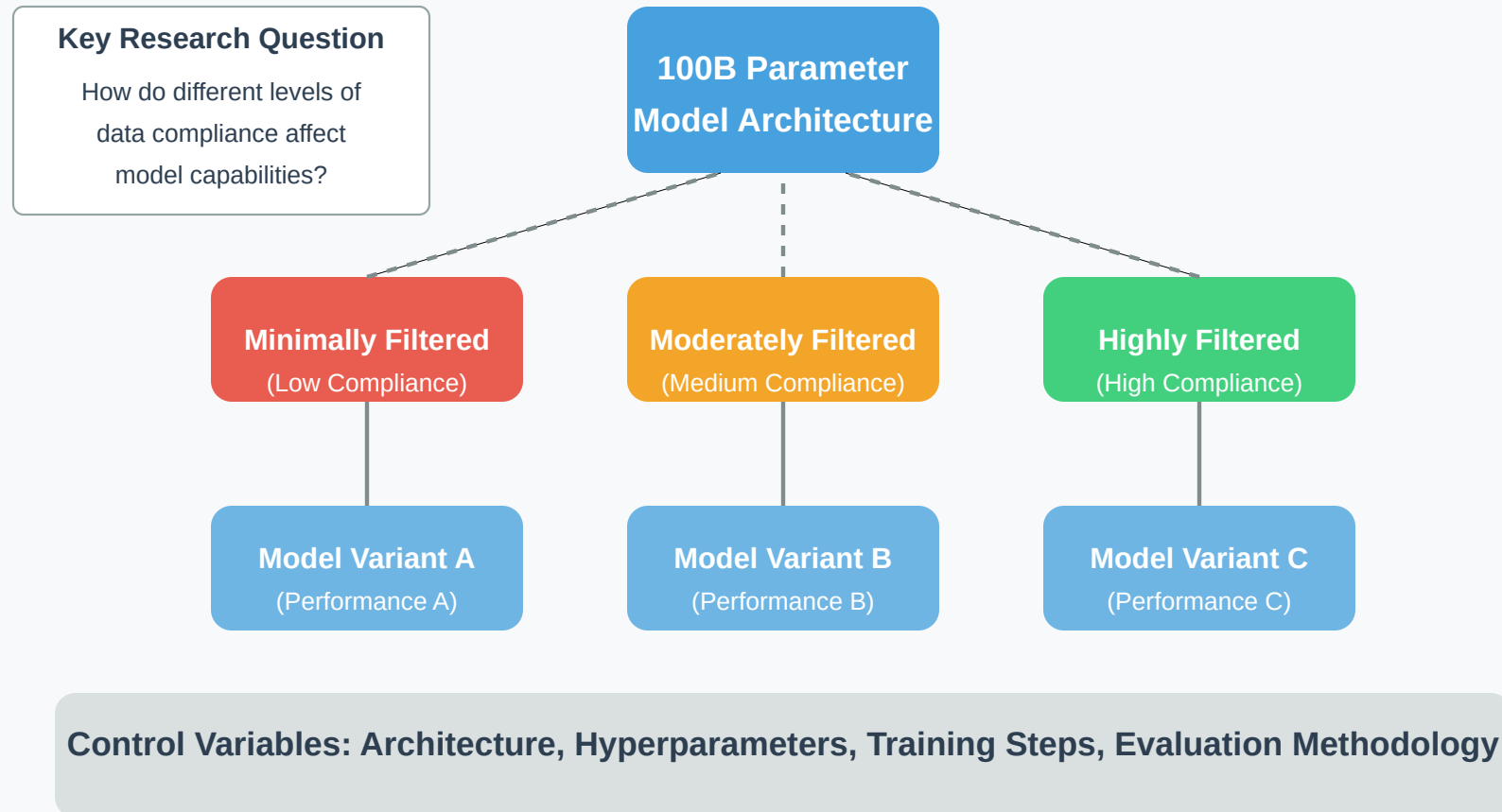


Complementary Nature:

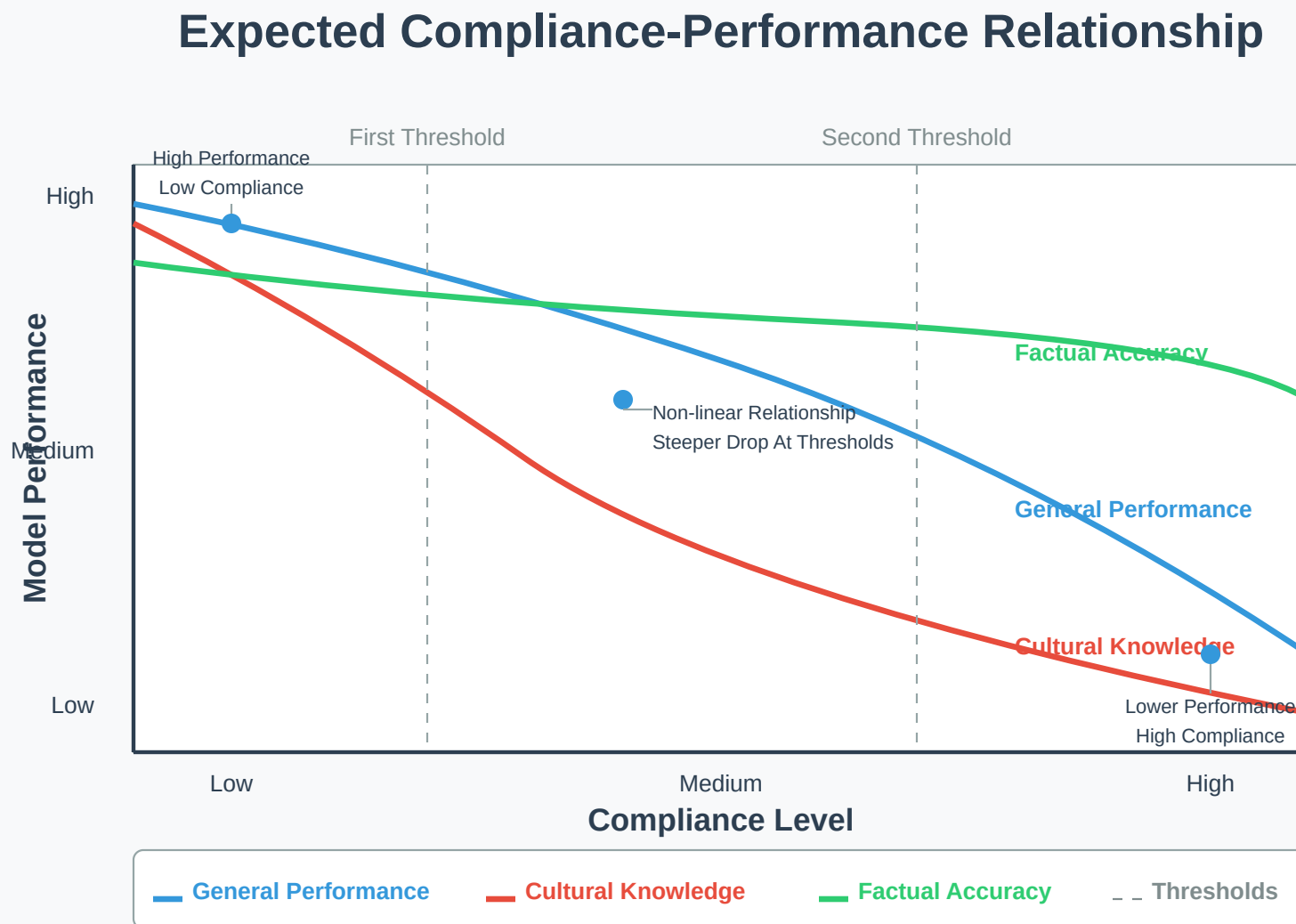
- Findings from both tracks inform each other
- Enables comprehensive understanding of both compliance and efficiency dimensions

Compliance vs. Performance Study (Leonardo)

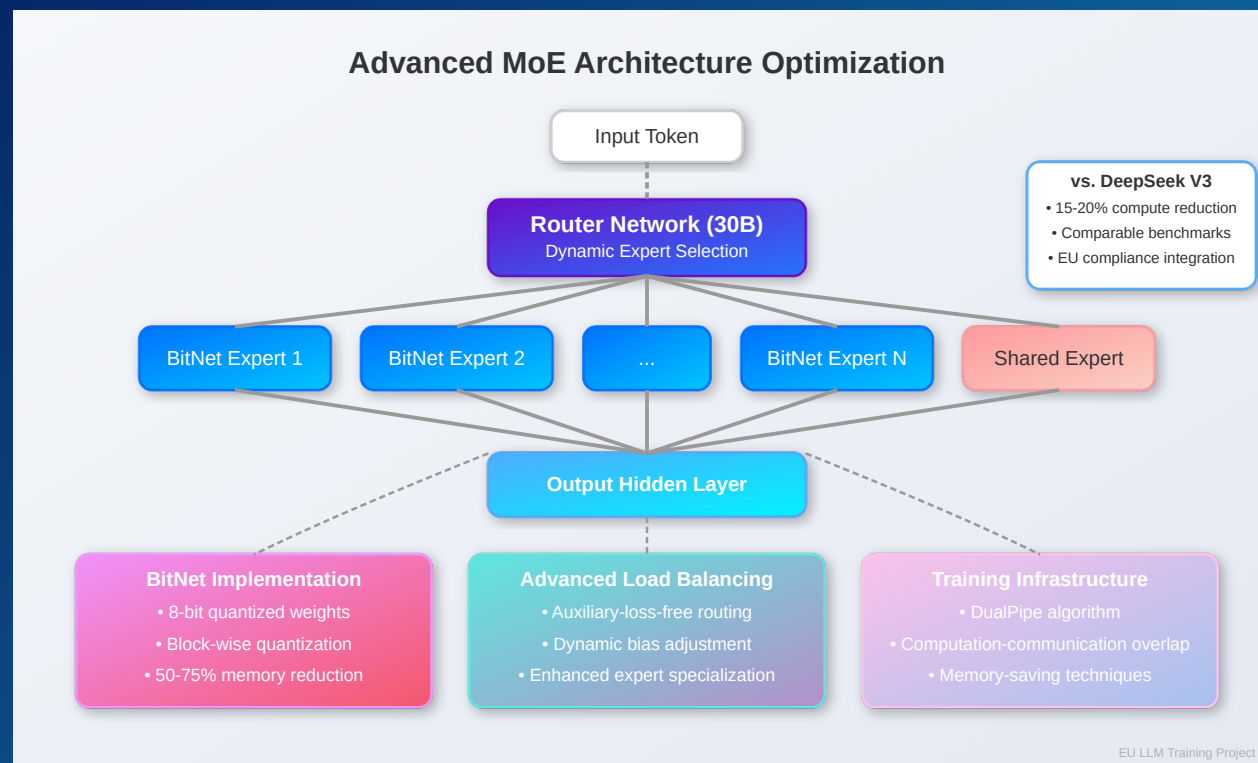
Compliance vs. Performance Study (Leonardo)



Expected Compliance-Performance Relationship



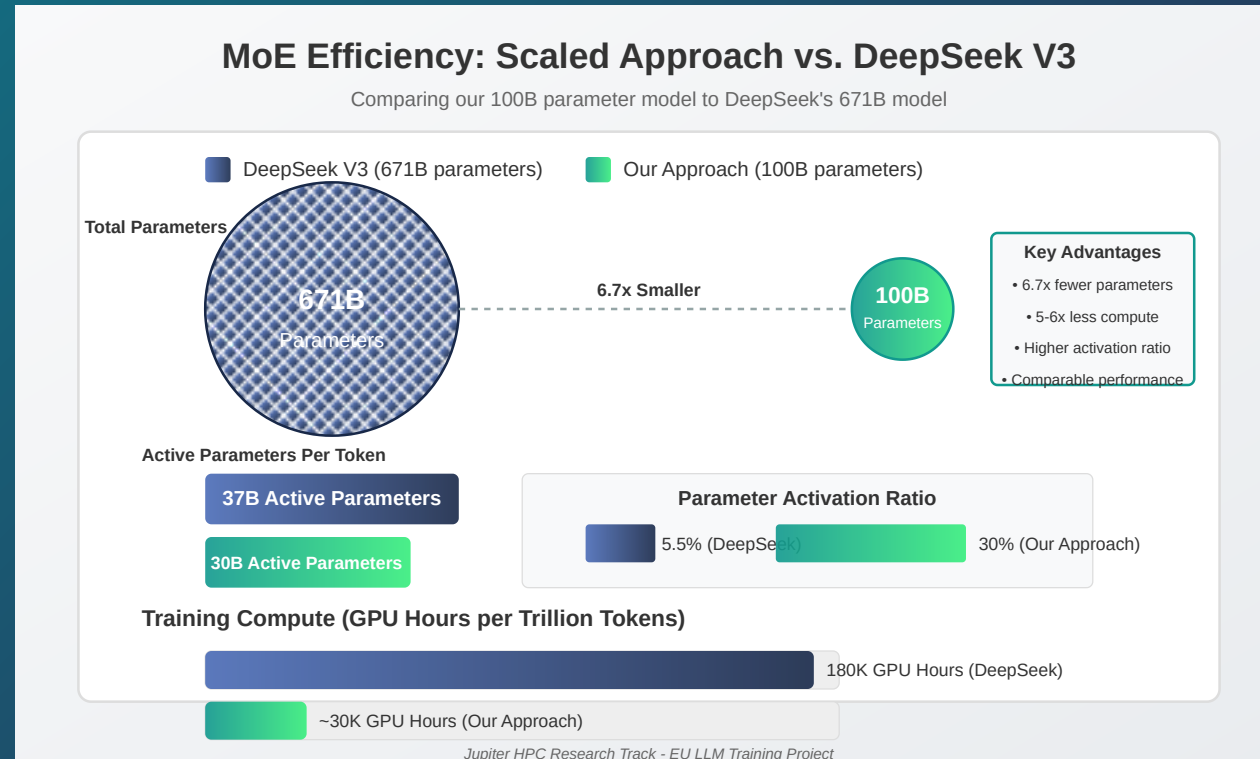
Advanced MoE Optimization Research (Jupiter)



Architecture Overview

- 30B parameter router with distributed experts
- Designed for efficient training on Jupiter's parallel computing architecture
- Leveraging sparse architecture principles from DeepSeekMoE

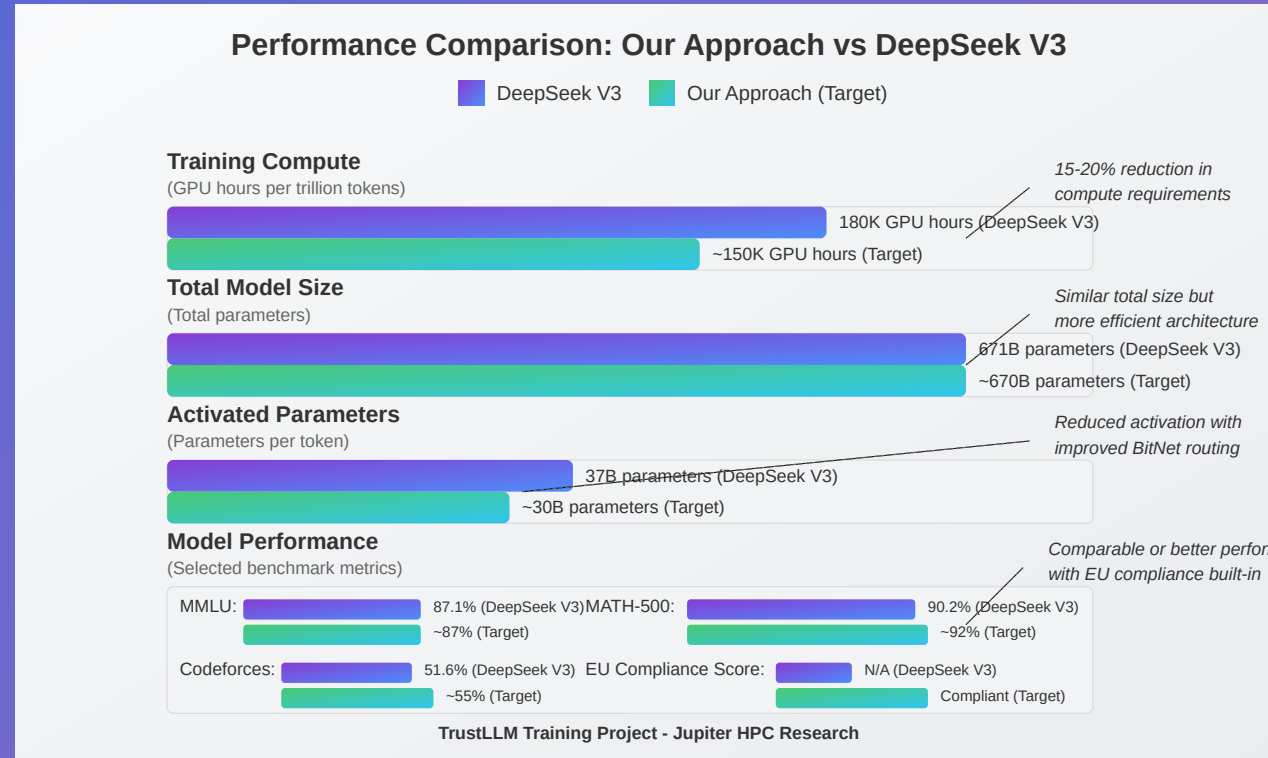
Performance Scaling Expectations



Parameter Activation Ratio Analysis

- DeepSeek: $671\text{B} \times 5.5\% = 36.9\text{B}$ effective parameters
- Our model: $100\text{B} \times 30\% = 30\text{B}$ effective parameters
- Similar performance with fewer effective parameters

Target: Matching DeepSeek V3 Performance



Long-term Performance Goals

- While our initial 100B model offers efficiency advantages, our ultimate goal is to match or exceed DeepSeek V3's performance
- Future iterations will aim to close any performance gaps while maintaining our compliance advantages

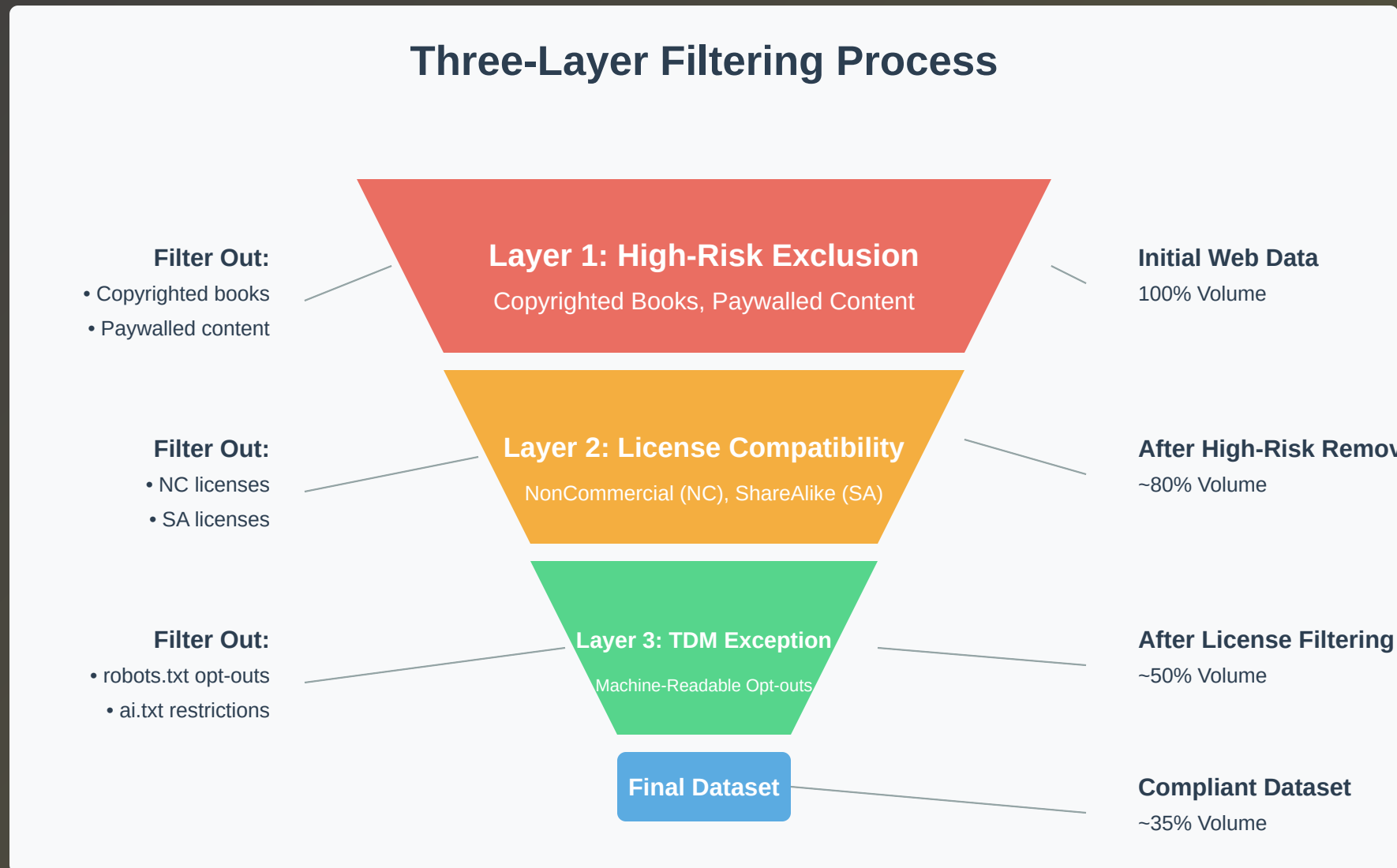
Data Compliance Approach



Implementation Considerations

- Balance between thorough filtering and preserving data utility
- Varying approaches for different languages and domains
- Integration with training pipeline for consistent application

Three-Layer Filtering Process



PII Detection Implementation

Structured Identifiers Targeted

- National ID numbers (format varies by country)
- Passport numbers
- Driver's license numbers
- Credit card numbers and CVV codes
- IBANs and bank account numbers
- IP addresses

Technical Approach

- High-precision regex patterns for each identifier type
- Language-specific pattern libraries
- Conservative matching criteria to minimize false positives

```
Swedish ID: ^\d{6}[-\s]\d{4}$  
IBAN: [A-Z]{2}\d{2}[\s](?:\d{4}[\s]){4}\d{2}
```

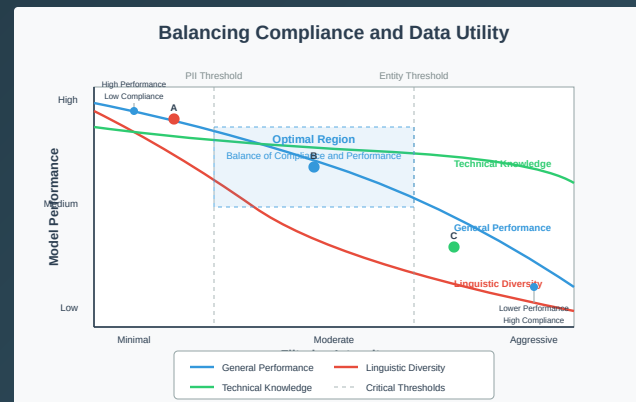
Balancing Compliance and Data Utility

Key Considerations

- Overly aggressive filtering risks removing valuable linguistic patterns
- Insufficient filtering creates regulatory and ethical issues
- Language-specific challenges require tailored approaches

Proposed Balance

- Focus on high-risk, clearly identifiable elements
- Preserve names, general locations, and common entities
- Document filtering decisions for transparency
- Multiple model variants with different filtering levels



EU Compliance Integration

Key Compliance Features

- Training data filtering for regulatory alignment
- Structured evaluation for bias and fairness metrics
- Documentation of model development process
- Risk assessment protocols integrated into evaluation framework
- Performance impact analysis at varying compliance levels

Target Outcome

- EU compliance without significant performance degradation
- Transparent model development process
- Documented tradeoffs between performance and compliance

Expected Outcomes and Significance

Scientific Value

- Quantified understanding of compliance-performance relationship
- Insights into efficient training methodologies for sparse models
- Data filtering impact assessment across linguistic contexts

Commercial Implications

- Framework for compliance-aware model development
- Optimized training approaches for reduced computing costs
- Clear guidance on regulatory boundaries for model deployment

Broader Impact

- Supporting EU's position in responsible AI development
- Advancing understanding of AI regulation implementation

Key Research Questions

Compliance Impact

- How do different levels of filtering affect model performance?
- Are there critical thresholds beyond which utility diminishes?
- Do different languages show varying sensitivity to filtering?

Technical Considerations

- Can we quantify the compliance-performance trade-off?
- How do filtering approaches affect downstream capabilities?
- Are there domain-specific impacts of different filtering levels?

Efficiency Analysis

- What are the most effective ways to optimize MoE architectures?
- How does our approach compare to established scaling laws?
- Can we achieve better parameter utilization than existing models?

Moving Forward Together

Immediate Action Items

- Finalize all technical preparations by April 14th, 2025 meeting
- Complete final infrastructure testing (May 2025)
- Prepare for June 20-30th, 2025 training start

Partner Collaboration

- WP2: Final data protocol validation
- Technical teams: Final infrastructure validation tests
- Evaluation team: Benchmark preparation

Feedback Request

- Input on additional compliance considerations
- Suggestions for evaluation metrics
- Approval of training timeline and resource allocation